

QMST5336: ANALYTICS

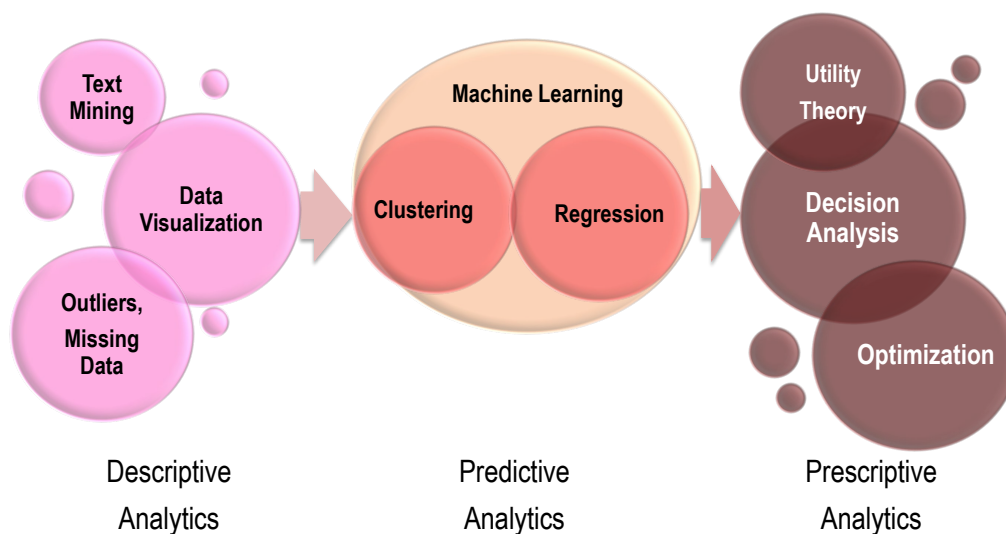
Lecture 8: Optimization

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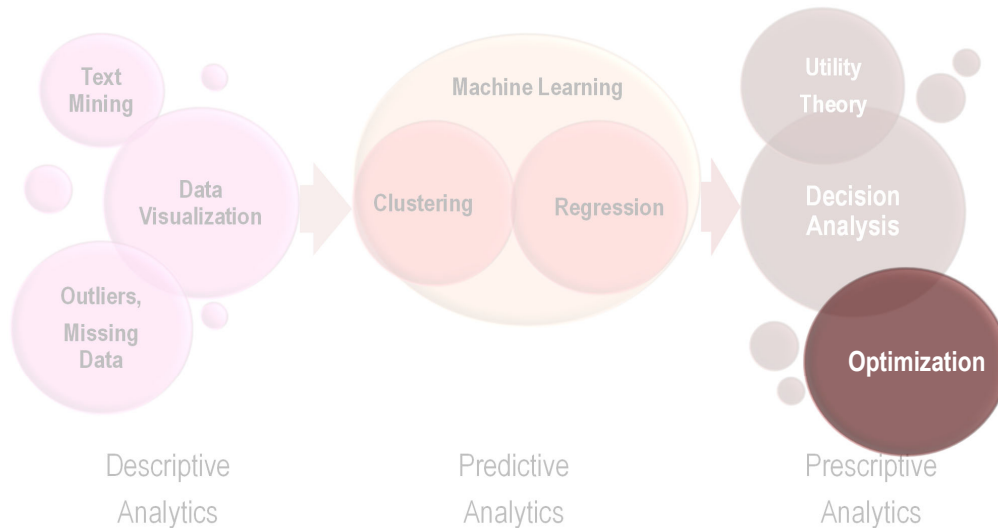
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Our Problem and Roadmap

An investment property in Austin: **Airbnb** versus **Rental**?



Roadmap: Where We Are



Case: One-off Utensils in the Airbnb Property

It is time consuming to clear and set up the house after every stay. In order to save cleaning times, we can use one-off utensils made in paper, plastic or bamboo. However, compared with ceramic/glass utensils, one-off ones are

- more costly
- not environment friendly and generating more garbage



Utensils	Labor(hr/stay)	Garbage(lb/stay)	Profit(\$/stay)
One-off	1	4	200
Ceramic	2	3	250

There are 20 hours of labor and 60 lbs of garbage allowance to spare per month.

Question 1: To maximize monthly profits, how many stays should we choose one-off utensils?

Linear Programming (LP) Model

Decision variables

- symbols representing levels or quantities of activity/actions

Objective function

- linear relationship of decision variables for the objective of a business
- most frequent business objective is to maximize profit
- most frequent objective of individual operational units (such as a production or packaging department) is to minimize cost

Constraint

- linear relationship representing a restriction on decision making

LP Model Formulation

- **Decision variables** x_j
- **Objective function** Max/Min $z = (c_1x_1 + c_2x_2 + \dots + c_nx_n)$
- **Constraints** subject to:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n (\leq, =, \geq) & b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n (\leq, =, \geq) & b_2 \\ & \dots \\ & \dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n (\leq, =, \geq) & b_n \end{cases}$$

where

- b_i are constraint levels,
- c_j are objective function coefficients,
- a_{ij} are constraint coefficients.

Case: LP Model Formulation

Utensils	Labor(hr/stay)	Garbage(lb/stay)	Profit(\$/stay)
One-off	1	4	200
Ceramic	2	3	250

There are 20 hours of labor and 60 lbs of garbage allowance to spare per month.

Decision variables

x_1 : the number of stays using one-off utensils per month;

x_2 : the number of stays using ceramic utensils per month.

Objective function Maximize $Z = \$200 x_1 + 250x_2$

Constraints Subject to

$$x_1 + 2x_2 \leq 20 \text{ hr}$$

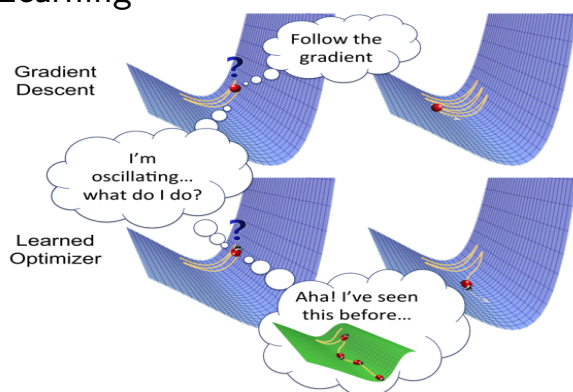
$$4x_1 + 3x_2 \leq 60 \text{ lb}$$

$$x_1, x_2 \geq 0$$

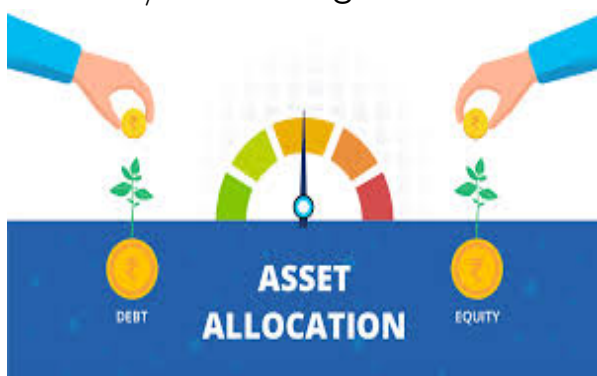
$$x_1 + x_2 \leq 30$$

Optimization Applications

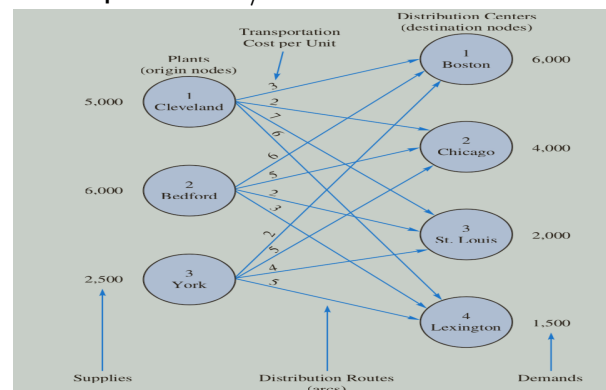
Artificial Intelligent & Machine Learning



Portfolio/Risk Management



Transportation/Distribution Network



Pricing/Revenue Management



Example 1: Formulate A Minimization Problem

M&D Chemicals produces two products that are sold as raw materials to companies that manufacture bath soaps and laundry detergents. Based on an analysis of current inventory levels and potential demand for the coming month, M&D's management specified that the combined production for products A and B must total at least 14,000 gallons. Separately, a major customer's order for 8,000 gallons consisting of product A and product B with ratio 1.25:1. M&D's objective is to satisfy these requirements at a minimum total production cost. Production costs are \$6 per gallon for product A and \$3 per gallon for product B.

Decision variables

x_1 : the amount of product A to produce in thousand gallons;

x_2 : the amount of product B to produce in thousand gallons;.

Objective function Minimize $Z = (000\$) 6x_1 + 3x_2$

Constraints Subject to

$$\begin{aligned} x_1 + x_2 &\geq 14 \\ x_1 &\geq 8 \times \frac{1.25}{2.25}, \quad x_2 \geq 8 \times \frac{1}{2.25} \end{aligned}$$

Example 2: LP Model Formulation

Round Tree Manor is a hotel that provides two types of rooms with three rental classes: Super Saver, Deluxe, and Business. The profit per night for each type of room and rental class is as follows:

	Rental Class (\$)		
Room	Super Saver	Deluxe	Business
Type I	30	35	-
Type II	20	30	40

Round Tree's management makes a forecast of the demand by rental class for each night in the future: 130 rentals in the Super Saver class, 60 in the Deluxe class, and 50 in the Business class. Round Tree has 100 Type I rooms and 120 Type II rooms. Can you develop a linear programming model to determine the amount of reservations to accept for each rental class in order to maximize profit per night?

Example 2: C'td

	Rental Class (\$)		
Room	Super Saver	Deluxe	Business
Type I	30	35	-
Type II	20	30	40

130 rentals in the Super Saver class, 60 in the Deluxe class, and 50 in the Business class. Round Tree has 100 Type I rooms and 120 Type II rooms.

Decision variables x_{1S} : the amount of rental class of room types;

Room	Super Saver	Deluxe	Business
Type I	x_{1S}	x_{1D}	-
Type II	x_{2S}	x_{2D}	x_{2B}

Objective function Maximize $Z = 30x_{1S} + 35x_{1D} + 20x_{2S} + 30x_{2D} + 40x_{2B}$

Constraints Subject to

$$x_{1S} + x_{1D} \leq 100$$

$$x_{2S} + x_{2D} + x_{2B} \leq 120$$

$$x_{1S} + x_{2S} \leq 130$$

$$x_{1D} + x_{2D} \leq 60$$

$$x_{2B} \leq 50$$

Investment Portfolio Selection Example 1/4

Portfolio selection problems involve situations in which a financial manager must select specific investments, for example, stocks and bonds, from a variety of investment alternatives.

Objective: Maximization of expected return or minimization of risk.

Constraints: Restrictions on the type of permissible investments, state laws, company policy, and so on.

Welte Mutual Funds, Inc., located in New York City, is looking for investment opportunities for \$100,000. The firm's top financial analyst identified five investment opportunities and projected their annual rates of return.

Investment	Projected Rate of Return (%)
Atlantic Oil	7.3
Pacific Oil	10.3
Midwest Steel	6.4
Huber Steel	7.5
Government bonds	4.5

Investment Portfolio Selection Example 2/4

Investment guidelines:

Neither industry (oil or steel) should receive more than \$50,000. Amount invested in government bonds should be at least 25% of the steel industry investments.

The investment in Pacific Oil, the high-return but high-risk investment, cannot be more than 60% of the total oil industry investment.

Define the following decision variables:

x_1 = dollars invested in Atlantic Oil

x_2 = dollars invested in Pacific Oil

x_3 = dollars invested in Midwest Steel

x_4 = dollars invested in Huber Steel

x_5 = dollars invested in Government bonds

Specify the objective: Maximizing return.

$$\max \quad 0.073x_1 + 0.103x_2 + 0.064x_3 + 0.075x_4 + 0.045x_5$$

Investment Portfolio Selection Example 3/4

Neither industry (oil or steel) should receive more than \$50,000. Amount invested in government bonds should be at least 25% of the steel industry investments.

The investment in Pacific Oil, the high-return but high-risk investment, cannot be more than 60% of the total oil industry investment.

Define the constraints:

Constraint 1: $x_1 + x_2 + x_3 + x_4 + x_5 = 100,000$

Constraint 2: $x_1 + x_2 \leq 50,000, x_3 + x_4 \leq 50,000$

Constraint 3: $x_5 \geq 0.25(x_3 + x_4)$

Constraint 4: $x_2 \leq 0.60(x_1 + x_2)$

Nonnegativity Constraint: $x_1, x_2, x_3, x_4, x_5 \geq 0$

Investment Portfolio Selection Example 4/4

Linear programming model for the Welte Mutual Funds investment problem:

$$\text{Max } 0.073x_1 + 0.103x_2 + 0.064x_3 + 0.075x_4 + 0.045x_5$$

s.t.

$$x_1 + x_2 + x_3 + x_4 + x_5 = 100,000 \quad \text{Available funds}$$

$$x_1 + x_2 \leq 50,000 \quad \text{Oil industry maximum}$$

$$x_3 + x_4 \leq 50,000 \quad \text{Steel industry maximum}$$

$$x_5 \geq 0.25(x_3 + x_4) \quad \text{Government bonds minimum}$$

$$x_2 \leq 0.60(x_1 + x_2) \quad \text{Pacific Oil restriction}$$

$$x_1, x_2, x_3, x_4, x_5 \geq 0$$

Marketing Example 1/5

Applications of linear programming to marketing include: advertising campaign mix, marketing mix, and marketing research. Consider the Business, Science, and Sports sections of a large content publisher, MHT. MHT has contracts for banner ads with five companies: Nile, Zstart, Cheetah, Stride, and Stove. Expected click-through rates for each advertiser by section of their web content is

Section	Nile	Zstar	Cheetah	Stride	Stove
Business	0.0155	0.0265	0.0100	0.0170	0.0105
Science	0.0165	0.0110	0.0125	0.0265	0.0125
Sports	0.0145	0.0235	0.0190	0.0225	0.0160

Contact Impression Limits

	Nile	Zstar	Cheetah	Stride	Stove
Lower Limit	1,500,000	1,000,000	1,100,000	1,000,000	1,500,000
Upper Limit	2,000,000	2,000,000	1,800,000	2,000,000	2,000,000

Marketing Example 2/5

For next Friday, MHT expects to have enough visitors to its website to guarantee 2,550,000 visitors to its Business section; 2,150,000 to its Science section; and 2,500,000 visitors to its Sports section. MHT receives \$0.30 for each click-through achieved. Advertising planners need to determine, for next Friday, how to allocate impressions for each of the five advertisers across the three sections of the MHT web site.

A linear programming model can be used to maximize revenue. The decision variables:

x_{ij} = the number of impressions in section i allocated to company j

where $i=1$ is Business $j=1$ is Nile
 $i=2$ is Science $j=2$ is Zstar
 $i=3$ is Sports $j=3$ is Cheetah
 $j=4$ is Stride
 $j=5$ is Stove

Marketing Example 3/5

The objective function is to maximize expected revenue from the three sections:

$$\begin{aligned} &0.3(0.0155x_{11} + 0.0265x_{12} + 0.01x_{13} + 0.017x_{14} + 0.0105x_{15}) \quad (\text{Business}) \\ &0.3(0.0165x_{21} + 0.011x_{22} + 0.0125x_{23} + 0.0265x_{24} + 0.0125x_{25}) \quad (\text{Science}) \\ &0.3(0.0145x_{31} + 0.0235x_{32} + 0.019x_{33} + 0.0225x_{34} + 0.016x_{35}) \quad (\text{Sports}) \end{aligned}$$

Maximize

$$\begin{aligned} &0.0047x_{11} + 0.008x_{12} + 0.003x_{13} + 0.0051x_{14} + 0.0032x_{15} + \\ &0.005x_{21} + 0.0033x_{22} + 0.0038x_{23} + 0.0008x_{24} + 0.0038x_{25} + \\ &0.0044x_{31} + 0.0071x_{32} + 0.0057x_{33} + 0.0068x_{34} + 0.0048x_{35} \end{aligned}$$

Marketing Example 4/5

Since each section is limited to the number of impressions that MHT can guarantee, "MHT expects to have enough visitors to its website to guarantee 2,550,000 visitors to its Business section; 2,150,000 to its Science section; and 2,500,000 visitors to its Sports section. "

So we have the following impression availability constraints:

$$x_{11} + x_{12} + x_{13} + x_{14} + x_{15} \leq 2,550,000 \text{ (Business Section Impression Available)}$$

$$x_{21} + x_{22} + x_{23} + x_{24} + x_{25} \leq 2,150,000 \text{ (Science Section Impression Available)}$$

$$x_{31} + x_{32} + x_{33} + x_{34} + x_{35} \leq 2,500,000 \text{ (Sports Section Impression Available)}$$

Marketing Example 5/5

Combining the previous information with a lower and upper limit on impressions received, we have the following linear program

$$\text{Maximize } 0.0047x_{11} + 0.008x_{12} + 0.003x_{13} + 0.0051x_{14} + 0.0032x_{15} +$$

$$0.005x_{21} + 0.0033x_{22} + 0.0038x_{23} + 0.0008x_{24} + 0.0038x_{25} +$$

$$0.0044x_{31} + 0.0071x_{32} + 0.0057x_{33} + 0.0068x_{34} + 0.0048x_{35}$$

$$x_{11} + x_{12} + x_{13} + x_{14} + x_{15} \leq 2,550,000 \text{ (Business Section Impression Available)}$$

$$x_{21} + x_{22} + x_{23} + x_{24} + x_{25} \leq 2,150,000 \text{ (Science Section Impression Available)}$$

$$x_{31} + x_{32} + x_{33} + x_{34} + x_{35} \leq 2,500,000 \text{ (Sports Section Impression Available)}$$

$$x_{11} + x_{21} + x_{31} \geq 1,500,000 \text{ (Lower Limit for Nile)}$$

$$x_{12} + x_{22} + x_{32} \geq 1,000,000 \text{ (Lower Limit for Zstar)}$$

$$x_{13} + x_{23} + x_{33} \geq 1,100,000 \text{ (Lower Limit for Cheetah)}$$

$$x_{14} + x_{24} + x_{34} \geq 1,000,000 \text{ (Lower Limit for Stride)}$$

$$x_{15} + x_{25} + x_{35} \geq 1,500,000 \text{ (Lower Limit for Stove)}$$

$$x_{11} + x_{21} + x_{31} \leq 2,000,000 \text{ (Upper Limit for Nile)}$$

$$x_{12} + x_{22} + x_{32} \leq 2,000,000 \text{ (Upper Limit for Zstar)}$$

$$x_{13} + x_{23} + x_{33} \leq 1,800,000 \text{ (Upper Limit for Cheetah)}$$

$$x_{14} + x_{24} + x_{34} \leq 2,000,000 \text{ (Upper Limit for Stride)}$$

$$x_{15} + x_{25} + x_{35} \leq 2,000,000 \text{ (Upper Limit for Stove)}$$

$$X_{ij} \geq 0 \quad i=1,2,3 \quad j=1,2,3,4,5$$

Integer linear programs

Problems that are modeled as linear programs with the additional requirement that one or more variables must be integer.

Eastborne Realty has \$2 million available for the purchase of new rental property. After an initial screening, Eastborne reduced the investment alternatives to townhouses and apartment buildings. Each townhouse can be purchased for \$282,000: Five are available. Each apartment building can be purchased for \$400,000. The developer will construct as many buildings as Eastborne wants to purchase. Eastborne's property manager can devote up to 140 hours per month to these new properties:

- Each townhouse is expected to require 4 hours per month.
- Each apartment building is expected to require 40 hours per month

Annual cash flow, after deducting mortgage payments and operating expenses, estimated to be:

- \$10,000 per townhouse.
- \$15,000 per apartment building.

Eastborne's owner would like to determine the number of townhouses and the number of apartment buildings to purchase to maximize annual cash flow.

Integer Linear Programming Example

Defining the decision variables:

T = number of townhouses.

A = number of apartment buildings.

Objective function for cash flow (in thousands of dollars):

$$\text{Max } 10T + 15A$$

Three constraints must be satisfied:

$$282T + 400A \leq 2000 \quad \text{Funds available (\$1000s)}$$

$$4T + 40A \leq 140 \quad \text{Manager's time (hours)}$$

$$T \leq 5 \quad \text{Townhouses available}$$

The variables T and A must be nonnegative.

In addition, the purchase of a fractional number of townhouses and/or a fractional number of apartment buildings is unacceptable.

T and A must be integer.

Eastborne Realty Example

Model for the Eastborne Realty problem:

Min $10T + 15A$

s.t.

$$282T + 400A \leq 2000$$

$$4T + 40A \leq 140$$

$$T \leq 5$$

$$T, A \geq 0 \text{ and integer}$$

This is a linear all-integer program.

Binary Variables: Capital Budgeting

Capital budgeting problem: A binary integer programming problem that involves choosing which possible projects or activities provide the best investment return.

In a capital budgeting problem, the objective function is to maximize the net present value of the capital budgeting projects.

The Ice-Cold Refrigerator Company is considering investing in several projects that have varying capital requirements over the next four years. Faced with limited capital each year, management would like to select the most profitable projects that it can afford.

Project Net Present Value, Capital Requirements, and Available Capital for the Ice-Cold Refrigerator Company

	Plant Expan (\$)	Warehouse Expan (\$)	New Machine (\$)	New Pro. Research (\$)	Total Cap. Available (\$)
Present V.	90,000	40,000	10,000	37,000	
Y1 Cap	15,000	10,000	10,000	15,000	40,000
Y2 Cap	20,000	15,000		10,000	50,000
Y3 Cap	20,000	20,000		10,000	40,000
Y4 Cap	15,000	5,000	4,000	10,000	35,000

Binary Variables: Capital Budgeting 2

Let us define four binary decision variables:

$P = 1$ if the plant expansion project is accepted; 0 if rejected.

$W = 1$ if the warehouse expansion project is accepted; 0 if rejected.

$M = 1$ if the new machinery project is accepted; 0 if rejected.

$R = 1$ if the new product research project is accepted; 0 if rejected.

This problem has four constraints:

One for the funds available in each of the next four years.

A binary integer linear programming model (thousands of dollars):

Max $90P + 40W + 10M + 37R$

s.t.

$15P + 10W + 10M + 15R \leq 40$ (Year 1 capital available)

$20P + 15W + 5R \leq 50$ (Year 2 capital available)

$20P + 20W + 10R \leq 40$ (Year 3 capital available)

$15P + 5W + 4M + 10R \leq 35$ (Year 4 capital available)

$P, W, M, R = 0, 1$

Product Design and Market Share Optimization 1/4

Conjoint analysis: A market research technique that can be used to learn how prospective buyers of a product value the product's attributes.

Product design and market share optimization problem: Sometimes called the share of choice problem, the choice of a product design that maximizes the number of consumers preferring it.

The results of conjoint analysis can be used in an integer programming model of a product design and market share optimization problem.

Part-worth: Utility value that a consumer attaches to each level of each attribute.

Salem Foods is planning to enter the frozen pizza market. Currently, two existing brands, Antonio's and King's, have the major share of the market. In trying to develop a sausage pizza that will capture a substantial share of the market, Salem determined that the four most important attributes when consumers purchase a frozen sausage pizza are crust, cheese, sauce, and sausage flavor.

In a typical conjoint analysis, a sample of consumers is asked to express their preference for a product with chosen levels for the attributes. Then, regression analysis is used to determine the part-worth for each of the attribute levels. The part-worths can be used to determine the overall value (utility) that each consumer attaches to a particular type of pizza. In general, each consumer's utility for a particular type of pizza is the sum of the part-worths for the attributes of that type of pizza.

Product Design and Market Share Optimization 2/4

Consumer	Thin Crust	Thick Crust	Mozzarella	Cheese Blend	Smooth Sauce	Chunky Sauce	Mild Sausage	Medium Sausage	Hot Sausage
1	11	2	6	7	3	17	26	27	8
2	11	7	15	17	16	26	14	1	10
3	7	5	8	14	16	7	29	16	19
4	13	20	20	17	17	14	25	29	10
5	2	8	6	11	30	20	15	5	12
6	12	17	11	9	2	30	22	12	20
7	9	19	12	16	16	25	30	23	19
8	5	9	4	14	23	16	16	30	3

Antonio's brand, with a thick crust, mozzarella cheese, chunky sauce and medium-flavored sausage is the current favorite for consumers 1, 4, 6, 7 and 8; King's brand, which has a thin crust, a cheese blend, smooth sauce and mild-flavored sausage, is the current favorite for consumer 2, 3, and 5.

Product Design and Market Share Optimization 3/4

Salem must design a pizza (choose the type of crust, cheese, sauce, and sausage flavor) that will have the highest utility for a sufficient number of people to ensure sufficient sales to justify making the product.

Assuming the sample of eight consumers in the current study is representative of the marketplace for frozen sausage pizza: Formulate and solve an integer programming model that can help Salem come up with such a design.

The decision variables are defined as:

$l_{ij} = 1$ if Salem chooses level i for attribute j ; 0 otherwise

$y_k = 1$ if consumer k chooses the Salem brand; 0 otherwise

Model for the Salem pizza problem:

Max $y_1 + y_2 + \dots + y_8$

$$11l_{11} + 2l_{21} + 6l_{12} + 7l_{22} + 3l_{13} + 17l_{23} + 26l_{14} + 27l_{24} + 8l_{34} \geq 1 + 52y_1$$

$$11l_{11} + 7l_{21} + 15l_{12} + 17l_{22} + 16l_{13} + 26l_{23} + 14l_{14} + 1l_{24} + 10l_{34} \geq 1 + 58y_2$$

$$7l_{11} + 5l_{21} + 8l_{12} + 14l_{22} + 16l_{13} + 7l_{23} + 29l_{14} + 16l_{24} + 19l_{34} \geq 1 + 66y_3$$

$$13l_{11} + 20l_{21} + 20l_{12} + 17l_{22} + 17l_{13} + 14l_{23} + 25l_{14} + 29l_{24} + 10l_{34} \geq 1 + 83y_4$$

$$2l_{11} + 8l_{21} + 6l_{12} + 11l_{22} + 30l_{13} + 20l_{23} + 15l_{14} + 5l_{24} + 12l_{34} \geq 1 + 58y_5$$

$$12l_{11} + 17l_{21} + 11l_{12} + 9l_{22} + 2l_{13} + 30l_{23} + 22l_{14} + 12l_{24} + 20l_{34} \geq 1 + 70y_6$$

$$9l_{11} + 19l_{21} + 12l_{12} + 16l_{22} + 16l_{13} + 25l_{23} + 30l_{14} + 23l_{24} + 19l_{34} \geq 1 + 79y_7$$

$$5l_{11} + 9l_{21} + 4l_{12} + 14l_{22} + 23l_{13} + 16l_{23} + 16l_{14} + 30l_{24} + 3l_{34} \geq 1 + 59y_8$$

Product Design and Market Share Optimization 4/4

Four more constraints ensure that one and only one level is selected for each attribute.

$$l_{11} + l_{21} = 1$$

$$l_{12} + l_{22} = 1$$

$$l_{13} + l_{23} = 1$$

$$l_{14} + l_{24} + l_{34} = 1$$

The optimal solution to this 17-variable, 12-constraint integer linear program:

$$l_{11} = l_{22} = l_{23} = l_{14} = 1 \text{ and } y_2 = y_5 = y_6 = y_7 = 1$$

The value of the optimal solution is 4:

Indicating that if Salem makes this type of pizza, it will be preferable to the current favorite for four of the eight consumers.

With $l_{11} = l_{22} = l_{23} = l_{14} = 1$, the pizza design that obtains the largest market share for Salem has a thin crust, a cheese blend, a chunky sauce, and mild-flavored sausage.

Note also that with $y_2 = y_5 = y_6 = y_7 = 1$, consumers 2, 5, 6 and 7 will prefer the Salem pizza.

Next Lecture

- **Solve** linear optimization problem in Python
- Sensitivity analysis of linear optimization
- Nonlinear optimization problem

Quiz 2 due on 11:59PM October 19.

